

Sentinel: Engineering Logbook

Build Narrative with Timestamps

For Samsung Solve for Tomorrow IRIS ISEF

Sentinel ? Engineering Logbook

Pre-Git ? Hardware Prototyping

Late 2025

Started with a dream: build a custom smart ring for emotion detection. Ordered MAX30102 PPG sensors. Prototyped on Arduino Nano. Got heart rate readings after a week of I2C register hacking.

Early 2026

HRV was a nightmare on raw sensor data. Peak detection, inter-beat intervals, frequency-domain analysis ? spent weeks getting a rolling 5-minute HRV window that didn't crash.

Jan 2026

Designed a custom 2-layer PCB in KiCad 7.0 ? nRF52840 BLE + MAX30102, 37 vias. Sent to JLCPCB (?240, 12-day lead).

Failure: PCB arrived. Soldered it. nRF52840 wouldn't flash.

`

JLinkARM.dll: Could not connect to target

`

Root cause: USB D+/D- traces were reversed. Took a week to trace the schematic error.

Feb 2026

OEM Pivot: Did the math. PCB rev2 = ?240 + 12 days. Enclosure molding = ?15,000 minimum. Battery certification = months. Time = 3+ months.

Decision: Stop building hardware. Use consumer smart rings (Oura, Ultrahuman, Apple Watch). Sentinel becomes a software platform. Any device exposing HR/HRV via API is compatible.

Phase 1 ? Streamlit Prototype

2026-03 to 2026-04

Built entire app in Streamlit 1.28. Monolithic architecture:

- main_.py (444 lines) ? app entry point
- agent_.py (720 lines) ? AI agent logic
- database.py (966 lines) ? raw SQLite
- crisis_.py ? crisis detection with timer
- patient_portal_.py, psychologist_.py ? dual portals
- ring_.py ? ring data simulator
- models/emotion_tfidf.pkl (4 MB) ? GoEmotions TF-IDF model

Bug: Streamlit single-threaded. time.sleep(3) auto-refresh for crisis alerts locked the entire app. Patient typing journal? Frozen.

2026-05-12 ? First git commit

5801c0e ? Repository initialized. Dependencies added, .venv removed, Groq API logging fixed.

2026-06-09 ? 16:45

e180f34 ? V1 ecosystem committed. Patient/psych portals, crisis detection, ring pipeline, AI agents, activity feed, triage, onboarding.

2026-06-16

UI sprint ? 9 commits:

- Calendar toggle bug (useEffect dependency)
- 3 palette iterations: dusk-slate ? rose-mauve ? warm neutrals
- 4 agent tests fixed
- Date-picker ? dropdown
- Mood per-day fix (UTC vs IST)
- Emoji visibility fix (CSS)

2026-06-19 ? 21:21

Onboarding flow, assigned psychs, TF-IDF emotion classifier training pipeline.

2026-06-20

Docs updated. Proposals finalized. Mood timeline chart + risk badge. Passwords fixed.

Phase 2 ? FastAPI + React Rewrite

2026-07-15

06:10 ? 14c1a01

Complete rewrite:

- Backend: FastAPI 0.109 + SQLAlchemy 2.0 + SQLite/WAL
- Frontend: React 19 + TypeScript 5.4 + Vite 6
- Auth: bcrypt + HS256 JWT + PBKDF2 unlock
- Docker Compose
- Thread-based crisis + WebSocket
- Rule-based discrepancy (18 positive, 22 negative words)
- 3-tier AI: Ollama ? Groq ? rule fallback
- 45 benchmarks across 5 categories

06:42 ? dc66736

SHA-256 hash-chained audit log with prev_hash.

07:21 ? Docker crash

`

sqlalchemy.exc.OperationalError: (sqlite3.OperationalError) unable to open database file

`

Fix: mkdir -p in Dockerfile + os.makedirs() in main.py.

18:51 ? 4cd42ec

Render deployment. Nginx WebSocket proxy.

Bug (3 hours): Nginx strips Upgrade header. Returns HTTP 200 instead of 101.

Fix:

`nginx

```
proxy_set_header Upgrade $http_upgrade;
proxy_set_header Connection "upgrade";
`
```

Phase 3 ? Security Hardening

2026-07-18 ? 05:00

Penetration test ? 22 findings: 4 critical, 8 high, 7 medium, 3 low.

08:00 ? 10:45 ? All 22 patched

A?J (10 original fixes)

| ID | Finding | Fix |

A	WebSocket no auth	get_current_user on WS
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| B | /crisis/* endpoints open | get_current_user on all 3 |

| C | Role escalation | Literal["patient","psychologist"] |

| D | No brute-force | failed_attempts + 15min lockout |

| E | No password policy | min_length=8, max_length=128 |

| F | IDOR | _owns_or_psych() guard |

| G | HMAC timing side-channel | hmac.compare_digest() |

| H | Root containers | USER sentinel in Dockerfiles |

| I | CORS wildcard | Explicit methods/headers |

| J | Default JWT secret | logger.warning() on startup |

K ? HttpOnly Cookie Auth

JWT was localStorage-only. Set HttpOnly cookie on login (httponly=True, samesite=lax, max_age=28800). get_current_user reads cookie fallback. /auth/logout clears it.

L ? EncryptedText TypeDecorator

Plaintext DB. EncryptedText(TypeDecorator) ? Fernet enc/dec transparent. Applied to 5 fields across 3 models. PBKDF2 600K + HKDF.

M ? RateLimiterMiddleware

No rate limiting. 100 req/min/IP. /health exempt. 429 + Retry-After.

N ? DOMPurify Sanitization

Raw text rendered unsanitized. DOMPurify(ALLOWED_TAGS=[], ALLOWED_ATTR=[]) on all text components.

O ? Internal Docker Network

Backend ports exposed to LAN. Removed ports:, internal: true network. Backend only via Nginx.

P ? Global Exception Handlers

Stack traces in 500s. Logged server-side. {"detail": "Internal server error"} to client.

Q ? Negation Handling

"I am not happy" ? false positive. 26 negation prefixes + 4-word window stripping. neutral+low_stress no longer alerts. Accuracy: 92% ? 96%. FN: 2 ? 0.

R ? Database Backup

backup_wal() copies .db + .db-wal to data/backups/ on every connection.

S ? Ollama Request Queuing

Multiple concurrent calls overwhelmed 8GB RAM. threading.Lock + 500ms gap. Rule fallback at 50ms.

10:45 ? Cleanup

Deleted software/ (59 files, ~10,700 lines Streamlit) ? archive/. Removed debug_imports.py.

Benchmarks

| Metric | Result |

Tests	43/45 pass
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| Accuracy | 96% (TP:21, TN:27, FP:2, FN:0) |

| SQLite | 48ms |

| Crypto 600K | 233.2ms |

Phase 4 ? Documentation

2026-07-18 ? 10:00 to 10:30

Updated 3 existing docs with all 22 findings:

TECHNICAL_DESIGN.md ? 14 edits. HttpOnly auth, negation handling, security section expanded 10?22, benchmarks updated.

JUDGE_QA.md ? 12 new Q&A entries across 6 categories (Encryption, Auth, Rate Limiting, Sanitization, Error Handling, Architecture).

ENGINEERING_DECISIONS.md ? 12 new rows (31?42) covering every fix.

Bug found: Agent wrote rows 31?42 with 5 columns instead of matching the 6-column format. Manually rewrote all 12 rows. Also fixed footer: "10 patches" ? "22 patches".

10:30 ? 11:30 ? Logbook + PDF

Created and revised ENGINEERING_LOGBOOK.md multiple times:

- v1: Full narrative ? "too casual"
- v2: First-person ? "make it specific"
- v3: Timestamps ? "include everything from start"

PDF generation ? 6 attempts:

| # | Error | Fix |

1	UnicodeEncodingException (Courier can't render Unicode)	Consolidate
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| 2 | Not enough horizontal space | Explicit set_x(14) |

| 3 | Same on long strings | 120-char pre-wrap |

| 4 | Same after page break | avail < 10 guard |

| 5 | Success ? 75 pages | Concise rewrite |

| 6 | Success ? 4 pages | Stable |

11:30 ? Session end

Phase 5 ? Crisis Escalation, PWA, Funding & Hardware (M0)

2026-07-19 ? Crisis escalation + trusted contact portal

- Journal entry with suicidal ideation ? classifier (high confidence) ? immediate action: suicide hotline card + trusted contacts page exposed on the patient dashboard.
- Trusted contact portal records who is allowed to see an escalation and verifies reachability.
- Real SMTP email sent to the trusted contact (mom@sentinel.demo) when a crisis is flagged; demo page set to idle (active: false) so no constant beep.
- Crisis for demo patient alaya handled end-to-end; trustee portal ADDRESSES gained an alaya entry.

2026-07-20 ? AI companion mode + PWA

- AI summarization now emits two outputs from the same journal entry: a warm, non-clinical companion summary for the patient and a structured OAP clinical summary for the psychologist.
- Frontend converted to an installable PWA: web manifest, icons, service worker with network-first API caching and stale-while-revalidate assets, offline app shell.

2026-07-31 ? Funding + hardware

- **Funding:**** Emergent Ventures grant received to support hardware procurement and pilot deployment.
- **Hardware:**** OEM smart ring sourced from Jport (China); vendor SDK/API + BLE interface (spec sheet pending from Jport).

2026-08-03 ? Hardware M0: ring ingestion + device binding

Implemented the ring SDK and secured device-binding layer, verified 9/9 SDK tests + end-to-end:

- **Model:**** RingDevice (ring_devices table) ? serial, owner patient FK, token_hash (SHA-256), revoked flag, last_seen_at.
- **API:**** POST /ring/pair (issues one-time device token; re-pairs revoked serials), POST /ring/unpair, GET /ring/devices (patient or psych), POST /ring/data (device-token or patient JWT).
- **Auth:**** get_ring_identity resolves device serial + token headers (or JWT fallback); constant-time compare, 401 on wrong/unknown/revoked.
- **SDK (app/services/ring):**** RingSource base + canonical SensorData payload; SimulatedRing (deterministic per-user-per-hour, calm/balanced/stressed), VendorAPIRingSource (vendor SDK/cloud _fetch hook), BLEGATTRingSource (bleak, HRM 0x2A37 parse, battery 0x180F, configurable char maps).
- **Scripts:**** sim_ring.py (device-token streaming, --once), ring_bridge.py (any source ? /ring/data, auto-pair), test_ring_api.py (9 tests incl. BLE parsers + missing-bleak guard).
- **Docs:**** docs/ROADMAP_HARDWARE.md created (M1?M3, Jport spec request sheet, architecture).
- **Demo:**** RING-DEMO-001 paired to alaya; verified token-auth pushes (id=13, id=14) and revocation ? re-pair flow.

Final Stack

Component Version	
Python	3.11
FastAPI 0.109	
SQLAlchemy 2.0	
React 19	
TypeScript 5.4	
Vite 6	
Docker Compose 2.24	
Nginx 1.25	
bleak (bridge-only) ~0.22	

Stats

- Lines: ~25,000
- Commits: 23+
- Security patches: 22 + device-token auth
- Benchmarks: 54 (52 pass; 9 ring API/SDK tests)
- Open TODOs: ~12

Late 2025 ? 2026-08-03

Sentinel: On-Premises Psychophysiological Triage Node

Samsung Solve for Tomorrow ? IRIS ? ISEF 2026